Step 2b - H8 - Higher ore extraction yields

Tech baseline

Technologies used in the market

Fine grinding to liberation size (incumbent)

How it works: Ball/SAG mills grind ore until valuable minerals are liberated; recovery is then achieved by flotation, gravity or leaching.

How well it performs: Grinding ~40% of mine energy (US DOE); typical BWi 10-20 kWh/t; liberation typically requires P80 of 75-300 um, finer for low-grade/complex ores.

Limits: Random bulk breakage liberates poorly at coarse sizes; overgrinding creates slimes that depress flotation/gravity recovery — the trade-off the hypothesis relaxes.

Flotation + regrind circuits (incumbent recovery method)

How it works: Rougher/scavenger flotation with regrind of middlings to recover locked particles.

How well it performs: Typical sulphide Cu recovery ~85-90%, gold ~85-95%; regrind adds several kWh/t.

Limits: Cannot recover values locked at grain boundaries below grind size; recovery gains cost disproportionate energy and capital — microwave cracks along grain boundaries instead.

Roasting / pressure oxidation (refractory ores)

How it works: Thermal or autoclave oxidation of pyrite/arsenopyrite or carbonaceous matter to expose encapsulated gold before leaching.

How well it performs: Enables >90-98% recovery on double-refractory ores but is highly energy- and capex-intensive (autoclaves cost $100M+).

Limits: Only economic at large scale; high emissions. Microwave roasting achieved >98% Au recovery in lab at lower specific energy than conventional roasting — a substitute/add-on niche.

HPGR for heap-leach liberation

How it works: HPGR micro-cracking improves permeability and exposes mineral surfaces before heap leaching.

How well it performs: Documented leach-kinetics and recovery improvements on gold/copper heap leach; ~1.5-4 kWh/t for the HPGR stage.

Limits: Mechanical cracks are less selective than grain-boundary fracture; benefit limited to leachable ores — microwave targets the same micro-crack mechanism more selectively.

High-voltage pulse fragmentation (SELFRAG)

How it works: Sub-microsecond discharges fragment ore preferentially along grain boundaries while immersed in water.

How well it performs: ~1-3 kWh/t for pre-weakening; lab/plant testwork shows coarser liberation and cleaner mineral surfaces than mechanical comminution.

Limits: Low throughput, wet processing, niche adoption; microwave achieves comparable grain-boundary selectivity dry and continuously at demonstrated 10-150 t/h scale.

Microwave pre-treatment (state of the art — the hypothesis technology)

How it works: Selective microwave heating of high-dielectric-loss phases generates intergranular cracks at mineral grain boundaries, liberating values at coarser sizes before mechanical grinding.

How well it performs: Porphyry Cu ore at ~2 kWh/t: equivalent liberation at 40-70 um coarser grind; +~1% Cu flotation recovery or +30-60 um coarser grind at equal recovery. Free-milling gold ore: BWi -18.5%, gravity gold recovery 28%→40%. Refractory concentrates: leach recovery 74%→>90-98% in lab studies.

Limits: Gains depend on dielectric/mineralogical contrast between valuable and gangue phases; recovery uplift of ~1% on porphyry ores is near the detection limit of flotation testwork; no commercial installation yet.

Why our solution can be better

Incumbents buy liberation with energy — grinding finer until values are exposed — which simultaneously destroys recovery via slimes and locked middlings. Microwave pre-treatment fractures ore preferentially AT grain boundaries, so values are liberated ~40-70 um coarser: the same recovery at lower energy, less overgrinding, and better downstream flotation/gravity/leach performance. The economics are powerful because recovery gains monetise directly as metal revenue (a +1% recovery point is worth several $M/yr on a large concentrator) rather than merely saving cost, and lab evidence shows order-of-magnitude larger effects on free-milling gold (28%→40% gravity recovery) and refractory ores (74%→>90% leach recovery) — niches where the mechanism is strongest.

Key assumptions

• Equivalent mineral liberation achievable at ~40-70 um coarser grind after ~0.3-2 kWh/t microwave treatment (measured, Nottingham pilot + MLA analysis)

• ~+1% Cu flotation recovery demonstrated at pilot-relevant doses; statistically hard to confirm but worth ~$5-15M/yr on a large concentrator (measured, Batchelor et al.)

• Free-milling gold gravity recovery improved 28%→40% after microwave pre-treatment; BWi -18.5% (measured, lab scale)

• Refractory gold leach recovery 74%→>90-98% via microwave pre-treatment/roasting at lower specific energy than conventional roasting (measured, lab scale)

• Recovery benefit requires dielectric/textural contrast between valuable and gangue minerals; ores with unfavourable textures show modest liberation gains (measured — key ore-selection risk)

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